

CARLA-Twin: A Large-Scale Digital Twin Platform for Advanced Networking Research

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Abstract—Digital Twins (DT) create virtual replicas of physical entities, enabling large-scale simulations and advanced data analysis in the twin world (a.k.a. the virtual environment). This capability significantly improves decision-making in the physical world, with applications spanning transportation, manufacturing, training, healthcare, etc. The success of DT hinges on fast and reliable networking technologies, which ensure efficient data collection from the physical world and seamless execution of decisions back into it. However, academic research in DT networking often faces challenges due to the lack of access to large-scale physical facilities, such as cities or warehouses. To address this gap, we develop a DT framework—CARLA-Twin—to facilitate networking research, leveraging the popular CARLA simulator—a large-scale autonomous driving research platform. In particular, CARLA-Twin employs a dual-computer architecture, where one instance of CARLA represents the physical world and the other acts as its twin world. These instances are interconnected through a bidirectional communication module to facilitate interactions between the physical and twin worlds. To validate the effectiveness of CARLA-Twin, we conduct a case study of DT-assisted city traffic management. Performance is evaluated under three scenarios: (1) Without DT Assistance, (2) DT with Ideal Communication, and (3) DT with Defective Communication. Simulation results demonstrate that DT scenarios have better performance compared to non-DT ones. Specifically, the collision rate is reduced by 95% under ideal communication scenario, while the average vehicle speed is nearly 2.8 times faster. However, elevated packet loss rates and network latency may diminish the effectiveness of DT systems, showing the importance of advanced networks. We will open-source our implementations.

Index Terms—Digital Twins, Networking, CARLA, Traffic Management

I. INTRODUCTION

Digital Twins (DT) is an emerging technology that has attracted considerable attention in recent years, inspiring numerous innovative studies that offer practical solutions for real-world applications [1]–[11]. A DT system typically consists of three main components: physical entities, virtual representations, and bidirectional information feedback links. These components work synergistically to bridge the gap between the physical and twin worlds, allowing the twin world to provide a deeper understanding of the corresponding physical world [3], [5], [12]. This enables better analysis and decision-making across diverse applications spanning transportation, manufacturing, training, healthcare, etc. For example, in a DT-assisted

traffic scenario, the twin world can provide recommendations based on a global perspective, effectively expanding the visual range of the driver, which improves both the safety and efficiency of driving.

The success of DT depends on fast and reliable networking technologies that facilitate robust bidirectional information feedback, forming the core of the DT system and distinguishing it from traditional simulations [1]. However, unstable communication links can severely degrade DT performance. For instance, in DT-assisted traffic management, delayed information may cause the twin world to issue incorrect recommendations, potentially leading to traffic accidents.

Despite extensive research on DT systems, limited attention has been paid to the impact of communication conditions. Existing studies often assume ideal communication environments or rely on network protocol simulators to approximate real-world deployment. For instance, Wu et al. [13] present an optimized DT controller for networked Internet-of-Things (IoT) devices in a smart city. They, however, assume the network is always reliable without defective communication. Sommer et al. [14] integrate the traffic simulator, SUMO [15], and the network simulator, OMNeT++ [16], to analyze the effects of network protocols on the traffic. Although insightful, such approaches are not fully embedded within the DT paradigm. Furthermore, Wang et al. [17] propose a co-simulation platform integrating SUMO and CARLA [18] for DT applications with vehicles and pedestrians. However, the study does not address the impact of communication quality.

To fully realize the potential of DT, examining how communication quality affects their reliability and efficiency is essential, but such research requires large-scale, such as city- or region-wide implementations. Unfortunately, most academic researchers lack the resources to support such scene construction, leading them to focus on small-scale scenarios, such as individual vehicle control or lab-based environments [19]. Scaling up DT systems poses challenges, including limited access to controllable physical entities, difficulties in high-quality data extraction, privacy concerns with sensitive information, and evaluations of decision-making policy in the physical world [20]. These barriers have hindered the widespread adoption of DT in real-world applications, leaving a gap in accessible and user-friendly DT frameworks. Therefore, developing a comprehensive DT system involving accurate

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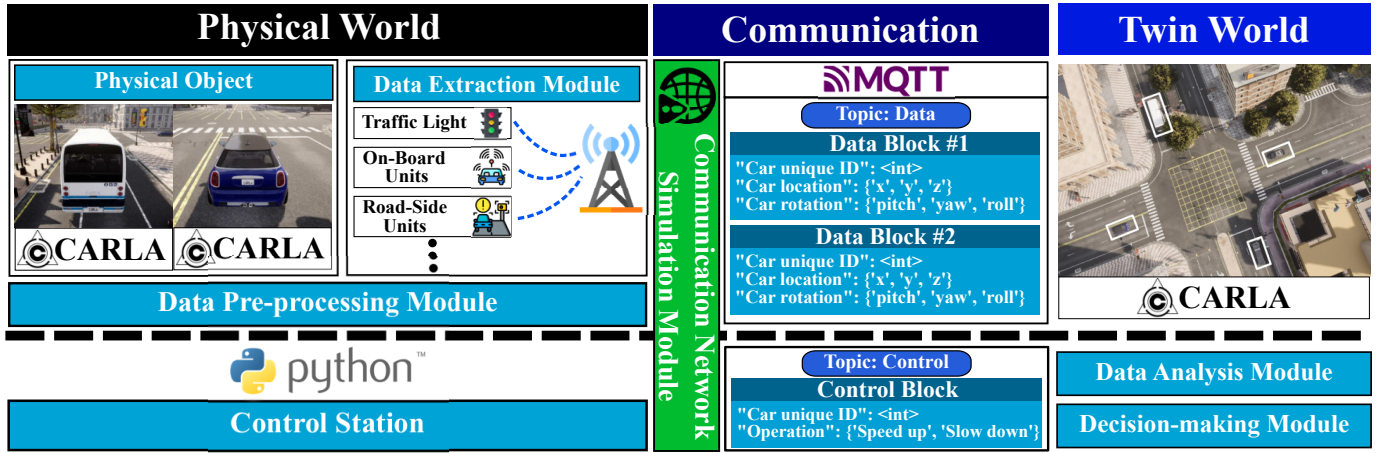


Fig. 1: Carla-Twin's system architecture.

data extraction from physical entities, virtual representation construction, seamless dynamic data transfer, and rigorous performance evaluations through verification, validation, and uncertainty quantification would greatly benefit integrated DT research and assessment [1].

To address these challenges, we present a user-friendly DT framework—CARLA-Twin—designed to investigate the impact of communication on DT-enabled applications. Our system is built on top of CARLA, an open-source simulator for large-scale autonomous driving research, with a highly realistic simulation environment with customizable features, such as sensor suits, vehicles, and urban landscapes [18]. In addition, CARLA supports the development, training, and validation of autonomous driving systems to enable controlled experiments in urban driving scenarios, providing robust metrics for performance analysis and facilitating safety assessments of complex and high-risk driving situations. Built on the developer version of CARLA, our system extends it by integrating a dedicated communication network simulation module. This module accurately models network performance issues, including network latency and packet loss, to evaluate their impacts on DT systems. Considering the small size of each data block in our system and the high frequency of updates, CARLA-Twin incorporates structured data transfer and an Message Queuing Telemetry Transport (MQTT)-based [21] messaging mechanism as bi-directional information feedback links. This design facilitates the analysis of network-induced challenges and their effects on overall system performance. In summary, our system serves as a comprehensive tool for investigating the interactions between communication and DT-enabled applications, providing valuable insights to advance the design and optimization of communication networks and protocols critical for DT systems.

To validate the effectiveness of our CARLA-Twin system, we investigate the impact of communication on DT-assisted city traffic management as a case study. Traffic management, an important part of smart traffic, can improve traffic efficiency and ensure traffic safety. Academia and industry have extensively explored methods to enhance vehicle perfor-

mance through assistive functionalities. With the advantages of DT technology, researchers have increasingly integrated DT concepts into smart traffic research, proposing a variety of innovative frameworks and definitions. For example, Wang et al. [8] employ DT technology to investigate signal-free intersections, utilizing collaborative vehicle perception to facilitate seamless navigation. Xu et al. [10] introduce and implement an advanced DT platform in Chattanooga (CTwin) to enhance urban mobility management. Wang et al. [11] present a smart mobility platform that incorporates vehicular technology with DT to enhance driving experiences. Wang et al. [4] develop a system to study the articulation problem of highway off-ramps by using the concept of DT and designing a driving user guidance interface. Despite these advancements, existing studies have largely overlooked the role of communication conditions in supporting the reliable operations of DT-enabled systems. In this paper, we evaluate the system's performance under three distinct operational conditions: (1) Without DT Assistance, (2) DT with Ideal Communication, and (3) DT with Defective Communication, offering insights into how communication conditions affect the functionality and reliability of DT. In summary, our main contributions include:

- An efficient, out-of-the-box DT framework that eliminates the need for data-gathering processes from the physical world.
- A detailed analysis of the impact of communication conditions on DT applications through DT-assisted city traffic management use case.
- A scalable platform that supports high-frequency real-time data feedback and structured storage for retrospective and predictive analysis.

II. CARLA-TWIN SYSTEM ARCHITECTURE

In this section, we introduce CARLA-Twin, which combines CARLA and MQTT to study the impact of networking on the DT system of a large-scale physical entity. CARLA-Twin consists of four core components: (1) CARLA-based Physical World, (2) Bidirectional Communication Module, (3) CARLA-based Twin World, and (4) Data Analysis, Simulation, and

Decision-Making, as shown in Fig. 1. The system involves two CARLA instances running concurrently: one representing the physical world, referred to as *CARLA-Phy*, and the other serving as the virtual twin world, called *CARLA-Vir*.

A. CARLA-based Physical World

The complexity of constructing real-world scenes and executing decisions back in the physical world makes conducting research in DT networking challenging and often impractical. To overcome these limitations, we utilize one CARLA platform to simulate the large-scale physical scenario called *CARLA-Phy*, allowing us to extract relevant data of interest and upload them to *CARLA-Vir*. For instance, specific vehicle information like location or speed can be extracted in a 'JSON' block format. Upon receiving the information from *CARLA-Phy*, *CARLA-Vir* generates an identical vehicle, mirroring the physical counterpart's state. This ensures the consistency between the physical and twin worlds, enabling accurate and efficient DT-based decision-making.

Enhancing Realism with Noise Modeling: Both physical and twin worlds operate under the same simulator, so we can achieve a difference-free DT scenario. However, real-world sensor data, such as vehicle sensor readings or traffic camera footage, often contain noise, and thus, practical DT may deviate from such exact replications. Therefore, to make the re-enacted scenes closer to real-world scenarios, we introduce appropriate noise and imperfection to the data extracted from *CARLA-Phy*. By doing so, our simulations can better mimic the inherent uncertainties present in real-world sensor data, thus enhancing the realism and applicability of our system. In this work, we add white Gaussian noise to the extracted data.

B. Bidirectional Communication Module

Compared to traditional simulations, a key advancement of DT is its capability for real-time bidirectional information feedback between the physical and virtual environments. Traditional simulations primarily focus on constructing virtual representations of the physical entities using detailed mathematical models and 3D modeling techniques [1]. Although these methods enable effective analysis and visualization, they are inherently limited by their lack of automated and dynamic feedback mechanisms to provide effective decision-making to maintain better physical world operations. In contrast, DT systems establish a continuous and synchronized flow of information, allowing real-time updates, monitoring, and reverse control. This enables DT to not only accurately reflect the current state of the physical world but also guide physical operations based on insights or analyses from the twin world, thereby fostering a closed-loop decision-making process. Such advancements are critical for applications requiring high precision, adaptability, and responsiveness, such as smart traffic, manufacturing, and infrastructure management.

To enable bidirectional information feedback of *CARLA-Twin*, we employ the MQTT protocol which is a lightweight publish-subscribe messaging protocol, specifically designed

for resource-constrained devices and networks with low bandwidth, high latency, or unreliable connections [21]. The MQTT communication protocol comprises three core components: the client (publisher or subscriber), the broker, and the topic. The broker serves as the central hub for information exchanges, receiving messages from publishers and routing them to subscribers based on predefined topics. Clients can dynamically publish or subscribe to broker messages using these topics that categorize data streams. For example, sensors can publish data on specific topics, such as "lidar" or "camera" depending on the type of sensor and the data being transmitted.

In our *CARLA-Twin* system, the size of data blocks is small (around 279 bytes), and the information update frequency is high due to the highly dynamic nature of vehicles. In addition, MQTT's flexibility further enables clients to act as both publishers and subscribers, ensuring seamless bidirectional communication between the physical and twin worlds. These characteristics make MQTT an ideal choice for our framework.

Communication Network Simulation Module: As a core system component, we develop a customized data processing layer to simulate real-world communication characteristics, including latency and packet loss. For example, we can make a probabilistic judgment at each data transmission to simulate the packet loss. By directly integrating this layer within the DT system, we eliminate the complexity of coupling CARLA with external platforms, resulting in a more streamlined, modular, and flexible solution. This integration enables precise control over simulated communication conditions, facilitating an in-depth analysis of how communication parameters impact critical aspects of DT operations, such as synchronization accuracy and confidence level of decision-making.

C. CARLA-based Twin World

In the twin world, maintaining all relevant physical entities is crucial for analysis, simulation, and decision-making. Upon receiving data from *CARLA-Phy*, *CARLA-Vir* evaluates whether the specified object exists within its environment. The synchronization of the twin world involves three primary tasks: object generation, object tracking, and object destruction.

During the operation of *CARLA-Twin*, the analysis module identifies the "Car ID" from each received data block. If this is the first instance of receiving information for a specific ID, the module executes the "object generation" function to initialize the vehicle. Otherwise, it updates the vehicle's position based on the received "object tracking" data. Additionally, if no data updates are received for a specific topic over an extended period, the system assumes that the vehicle has been destroyed and triggers the "object destruction" function.

In real-world scenarios, the focus is often limited to specific regions of interest, such as intersections or areas prone to accidents. Constructing a twin world with an infinite Field of View (FoV) is computationally infeasible. The twin world is also built based on data from specified zones, leveraging hardware such as roadside cameras and onboard units like in-vehicle LiDAR. When an object exits the designated analysis area, its information is no longer required. To manage this, our

system employs a time threshold for each object. If *CARLA-Vir* does not receive an update for an object before this threshold, it assumes the object left the area of interest and subsequently destroys its virtual representation. This ensures efficient use of computational resources while maintaining accurate and dynamic operations of *CARLA-Vir*.

D. Data Analysis, Simulation, and Decision Making

Our analysis tool leverages the real-time status and historical dynamics of each physical entity for decision-making or instructions. A database is established at the receiving end, where all incoming information is systematically organized by each entity's unique identifier (Car ID). Using timestamps associated with the data, the relative information (e.g., the location and speed of all vehicles) can be easily reproduced.

This collated data can be synthesized and analyzed in the *CARLA-Vir* to facilitate evaluating various management strategies. For example, for smart traffic management, several key performance indicators, such as vehicle waiting time and collision probability, can be evaluated to determine the most effective traffic planning scenarios. Based on these insights, appropriate suggestions are issued to the physical entities to optimize real-world traffic management.

III. CASE STUDY: CITY TRAFFIC MANAGEMENT

This section evaluates the impact of communication issues like latency and packet loss on DT-assisted applications with a case study—city traffic management. Through three different scenarios, (1) Without DT Assistance, (2) DT with Ideal Communication, and (3) DT with Defective Communication, our study highlights how communication defects can disrupt DT decision-making, leading to reduced average vehicle speed and increased collision rates. Results show that ideal communication significantly improves traffic performance—almost 2.8 times higher average vehicle speeds and reducing collisions by over 95% compared with “Without DT Assistance”—while even minor communication defects noticeably degrade outcomes, underscoring the importance of reliable, low-latency communication for effective DT deployment.

A. Impact of Wireless Communications on Transportation

In recent years, there has been a significant increase in research using DT to investigate transportation strategies and assisted driving systems. In such systems, seamless real-time data streaming ensures reliable decision-making. Fig. 2 illustrates two examples where communication performance has a significant impact on DT-guided vehicle operations:

- Intersection Merging:** The top of Fig. 2 shows an intersection merging situation managed by DT. In this case, transmission latencies cause the twin world to operate with outdated vehicle positions. Consequently, the DT issues unnecessary or wrong commands, such as allowing other vehicles to accelerate, which can lead to crashes.
- Off-ramp Merging:** The bottom of Fig. 2 demonstrates a situation where DT-assisted vehicles merge on an off-ramp. Here, the twin world issues a merge recommendation based on received data. However, due to signal

loss, the true position of the merging vehicle on the off-ramp cannot be updated accurately, leading to erroneous decision-making by the DT. This inaccuracy could result in a potential traffic accident.

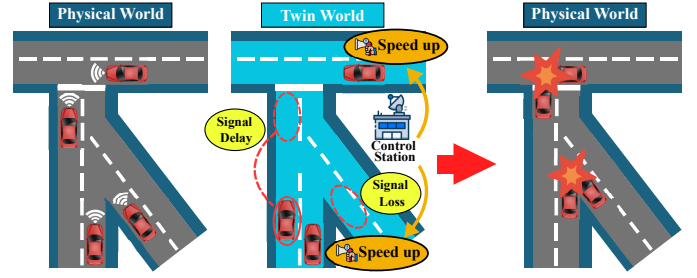


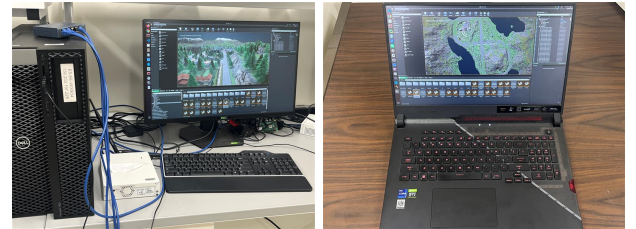
Fig. 2: Collision due to latency and packet loss.

These examples highlight the critical role of communication quality in DT-enabled transportation systems. Addressing these issues is essential to enhance the robustness and effectiveness of such DT-enabled applications, ensuring both traffic safety and efficiency.

To investigate the impact of communication on DT applications, this case study demonstrates the feasibility of the developed *CARLA-Twin* platform by evaluating the city traffic management scenario supported by DT. Specifically, we mainly examine the effects of network latency and packet loss on transportation performance. We employ two key metrics to assess the outcomes: average vehicle speed and collision rate. The average vehicle speed evaluates the experience from the drivers' perspective, capturing both efficiency and comfort. The collision rate serves as a measure of driving safety, reflecting the system's ability to prevent accidents.

B. Implementation and Experiment Setup

Hardware and Environment: The system consists of two servers for *CARLA-Phy* and *CARLA-Vir* deployment, respectively. *CARLA-Phy* is deployed on a workstation with Intel Xeon(R) Gold 6242R CPU @ 3.10GHz × 80, NVIDIA Corporation TU104GL [Quadro RTX 5000] × 2, 126 GB memory, 5 TB disk, and Ubuntu 20.04.6 LTS. *CARLA-Vir* is deployed on a laptop with 12th Gen Intel® Core™ i9-12900H × 20, NVIDIA GeForce RTX 3070 Ti Graphics Card, 32 GB memory, 2 TB disk, and Ubuntu 20.04.6 LTS. The platform of our system is shown in Fig. 3.



(a) Server for *CARLA-Phy*. (b) Server for *CARLA-Vir*.

Fig. 3: Experimental platform.

We deploy *CARLA-Twin* to investigate the impact of communications on DT-assisted city traffic management using the

built-in city model of CARLA, called “Town 04”. The model incorporates a variety of road types, including a network of small, short streets with multiple intersections, as well as an “8-shaped” ring road that encircles buildings and nearby mountainous terrain. These diverse features create a realistic and dynamic environment, making it well-suited for analyzing complex traffic scenarios.

We randomly generate multiple vehicles and assign start and end points for each physical vehicle, ensuring that the locations are near residential areas or at the edges of the map, with a minimum distance of at least one kilometer between them. The simulation does not have a fixed duration. After 40 driving tasks are completed, the collected data are analyzed for statistical insights. Specifically, we evaluate three scenarios under a data transmission frequency of 10 Hz:

- **Without DT Assistance:** Running *CARLA-Phy* alone and not taking the instructions from *CARLA-Vir*.
- **DT with Ideal Communication:** Using the *CARLA-Vir* instruction to assist the running *CARLA-Phy*, assuming perfect communication conditions with zero network latency and zero packet loss.
- **DT with Defective Communication:** *CARLA-Phy* is run with the aid of the *CARLA-Vir* instructions, and the Communication Network Simulation Module is enabled to manually set the Packet Loss Rate (PLR), which is the fraction of failed transmitted packets and the network latency to explore the system performance under different communication conditions.

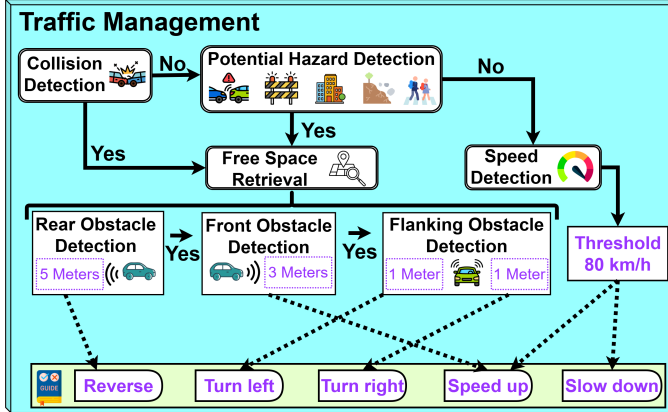


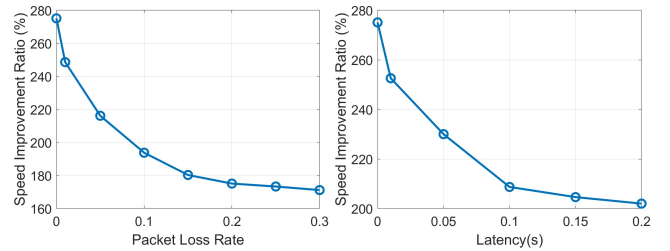
Fig. 4: Decision-making logic of traffic management.

DT Decision-making Logic: The focus of this paper is platform development rather than strategy design. Therefore, we employ an intuitive processing logic based on the information provided by *CARLA-Phy*. More specifically, rather than introducing complex or advanced traffic analysis and obstacle avoidance algorithms, we design an intuitive judgment logic for each vehicle based on the global FoV in the twin world. The main logic, as illustrated in Fig. 4, begins with the twin world checking for collisions. It then identifies potential risks for each vehicle, such as nearby obstacles (e.g., other vehicles, buildings, etc.). For example, if no collision is detected, no potential risks are identified, and the vehicle’s current

speed is below 80 km/h, the twin world issues a “speed up” instruction. Otherwise, the system determines and provides the most appropriate instruction based on the current conditions.

C. Performance Evaluations

In this subsection, we summarize the experimental results on transportation performance, focusing on the average speed and collision rate in three scenarios. The average speed for scenario “Without DT Assistance” is 23.66 km/h. We set this as the baseline and calculated the improvement ratio of average speed for the other two scenarios. From Fig. 5, we observe that the average vehicle speed in the “DT with Ideal Communication” scenario is nearly 2.8 times higher than that of the “Without DT Assistance” scenario. This significant improvement is attributed to the DT’s capability to provide globally optimized and reliable vehicle acceleration recommendations. Furthermore, the average speed improvement ratio decreases as network latency or packet loss rate increases. Interestingly, when packet loss or latency are small, their impact on DT performance is more pronounced, underscoring the critical need for reliable, low-latency communication to leverage the benefits of the DT system fully.



(a) Impact of packet loss rate.

(b) Impact of latency.

Fig. 5: Impact of packet loss and latency on the vehicle speed.

Next, we evaluate the vehicle collision performance. We use the reduction percentage in collision frequency as the performance metric, compared to the “Without DT Assistance” scenario, which results in almost 0.03 collisions per kilometer. As shown in Fig. 6, the “DT with Ideal Communication” scenario reduces collision by nearly 95%. While the “DT with Defective Communication” scenario shows a noticeable decline in collision reduction compared to the “DT with Ideal Communication” scenario, it still achieves a clear collision reduction compared to the “Without DT Assistance” scenario. Unlike the average speed performance, minor packet loss or latency has only a slight impact on collision reduction. This is expected as vehicle speed decreases more significantly in such cases, thereby reducing the likelihood of collisions.

Our experiments reveal that even with a 30% PLR, the DT system still enhances average speed with a reduced collision rate. This is due to the acceleration criteria using a fixed threshold without considering speed limitations. Despite the high PLR, the DT system achieved an update frequency of approximately 7 Hz per vehicle, demonstrating the superiority of global coordination over independent control. For network latency, our findings indicate that even as latency increases, the DT system still performs better than the “Without DT

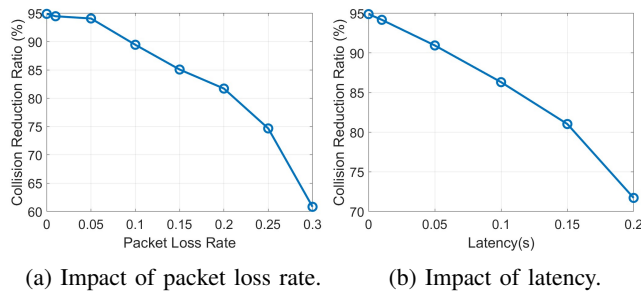


Fig. 6: Impact of packet loss and latency on the collision rate.

Assistance” scenario. We believe this is because all vehicles experience the same latency, maintaining a synchronized lag state. However, if vehicles were assigned varying latencies, the performance of the DT system would likely be degraded.

Rationalization of decision-making: The performance of a DT system is also heavily influenced by the quality of its data analysis and policy design. Although this aspect is beyond the scope of this work, it’s important to highlight that poorly designed algorithms can severely degrade DT performance, even resulting in outcomes worse than non-DT scenarios.

IV. CONCLUSION AND FUTURE DIRECTIONS

In this paper, we present CARLA-Twin, a platform designed to facilitate networking research for Digital Twins (DT) systems by leveraging the widely-used CARLA simulator to address the challenge of accessing large-scale physical facilities. CARLA-Twin consists of: (1) CARLA-based Physical World, (2) Bidirectional Communication Module, (3) CARLA-based Twin World, and (4) Data Analysis, Simulation, and Decision-Making. The platform also enables seamless configuration of communication modules, allowing researchers to study the impact of communication quality on DT performance. To validate the effectiveness of CARLA-Twin, we apply it to DT-assisted city traffic management as a case study. Results show that under ideal communication, the DT system can improve the average vehicle speed by nearly 2.8 times compared to the “Without DT Assistance” scenario, and reduce the collision rate by over 95%. However, imperfect communication significantly degrades system performance, highlighting the critical role of reliable and low-latency communication in DT systems.

In the future, we plan to integrate networking simulators, such as NS3 or OMNeT++, into our system to enable high-fidelity simulations of Vehicle-to-Everything (V2X) protocols, including Dedicated Short-Range Communications (DSRC) and Long Term Evolution (LTE)-V2X. This integration will allow for more diverse network topologies and experimental designs, better reflecting real-world deployment scenarios where multiple routers, base stations, and communication constraints influence system performance. Additionally, we aim to investigate the impact of network fluctuations, leading to heterogeneous network conditions across different vehicles.

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